CS:5980
Foundations of Embedded Systems

Liveness Requirements
Part II

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LTL Recap

Syntax: Formulas built from

- Base formulas: Boolean-valued expressions over typed variables
- Logical connectives: $\wedge$, $\vee$, $\Rightarrow$, $\neg$, …
- Temporal Operators: **Always**, **Eventually**, **Next**, **Until**

Semantics: defined by rules for the satisfaction relation

- Formulas are evaluated w.r.t. a trace ρ (infinite sequence of valuations)
- A system satisfies spec φ iff every infinite execution satisfies φ

Derived operators:

Repeatedly _	(**Always Eventually _**)
Persistently _	(**Eventually Always _**)

Sample requirement: Every request is eventually granted

Always [req = 1 $\Rightarrow$ **Eventually** (grant = 1)]

Temporal Implications and Equivalences

Understanding subtle differences among different variants of LTL formulas can be tricky

Definition: Let φ_1, φ_2 be LTL formulas

1. φ_1 is *stronger* than φ_2 if every trace that satisfies φ_1 satisfies φ_2 too
   - i.e., every trace satisfies the implication $\varphi_1 \Rightarrow \varphi_2$
2. φ_1 is equivalent to φ_2 if φ_1 and φ_2 are satisfied by exactly the same traces
   - i.e., each formula is stronger than the other
   - i.e., every trace satisfies the double implication $\varphi_1 \Leftrightarrow \varphi_2$
   - i.e., the two formulas express exactly the same requirement

Knowing some standard equivalences is useful for simplifying formulas

Temporal Implications and Equivalences

- **Always** φ is stronger than φ
- **Always** φ is equivalent to **¬Eventually** $\neg\varphi$
- **Repeatedly** φ is equivalent to **¬Persistently** $\neg\varphi$
- **Persistently** φ is stronger than **Repeatedly** φ
- **Always** φ is equivalent to $\varphi \wedge$ **Next Always** φ
- **Eventually** φ is equivalent to $\varphi \vee$ **Next Eventually** φ
- φ_1 **Until** φ_2 is equivalent to $\varphi_2 \vee [\varphi_1 \wedge$ **Next** $(\varphi_1$ **Until** $\varphi_2]$

Exercise: What is the mutual relationship between these formulas?

1. **Always Eventually** φ
2. **Next Always Eventually** φ
3. **Eventually Always Eventually** φ

Logical Connectives and Temporal Operators

Are these two formulas equivalent?

Eventually $(\varphi_1 \vee \varphi_2)$ $\qquad$ **Eventually** $\varphi_1 \vee$ **Eventually** φ_2

Yes, they are

Proof:

⇒) Suppose a trace ρ satisfies **Eventually** $(\varphi_1 \vee \varphi_2)$

- There is a position j such that $(\rho, j) \vDash \varphi_1 \vee \varphi_2$
- Either $(\rho, j) \vDash \varphi_1$ or $(\rho, j) \vDash \varphi_2$
- Suppose $(\rho, j) \vDash \varphi_1$ (the other case is similar)
- Then ρ satisfies **Eventually** φ_1
- Hence, it also satisfies **Eventually** $\varphi_1 \vee$ **Eventually** φ_2

⇐) Suppose a trace ρ satisfies **Eventually** $\varphi_1 \vee$ **Eventually** φ_2

- Suppose ρ satisfies **Eventually** φ_1 (the other case is similar)
- There is a position j such that $(\rho, j) \vDash \varphi_1$
- Then, $(\rho, j) \vDash \varphi_1 \vee \varphi_2$
- It follows that ρ satisfies **Eventually** $(\varphi_1 \vee \varphi_2)$

Logical Connectives and Temporal Operators

Are these two formulas equivalent?

Eventually $(\varphi_1 \wedge \varphi_2)$ **Eventually** $\varphi \wedge$ **Eventually** φ_2

The first is stronger than the second but not vice versa

Proof:

⇒) Suppose a trace ρ satisfies **Eventually** $(\varphi_1 \wedge \varphi_2)$

- There exists a position j such that $(\rho, j) \vDash \varphi_1 \wedge \varphi_2$
- It follows that both $(\rho, j) \vDash \varphi_1$ and $(\rho, j) \vDash \varphi_2$
- Since $(\rho, j) \vDash \varphi_1$ then ρ satisfies **Eventually** φ_1
- Similarly, it also satisfies **Eventually** φ_2
- It follows that ρ satisfies **Eventually** $\varphi_1 \wedge$ **Eventually** φ_2

⇐) To disprove this, consider trace 0,1,0,1,0,1,… over a Boolean variable x

- Trace satisfies **Eventually** $(x = 0) \wedge$ **Eventually** $(x = 1)$
- But does not satisfy **Eventually** $(x = 0 \wedge x = 1)$

Logical Connectives and Temporal Operators

Distributivity rules for logical connectives and temporal operators

Exercise: Are these pairs of formulas equivalent?

1. **Always** $(\varphi_1 \wedge \varphi_2)$ **Always** $\varphi_1 \wedge$ **Always** φ_2
2. **Always** $(\varphi_1 \vee \varphi_2)$ **Always** $\varphi_1 \vee$ **Always** φ_2
3. **Repeatedly** $(\varphi_1 \wedge \varphi_2)$ **Repeatedly** $\varphi_1 \wedge$ **Repeatedly** φ_2
4. **Repeatedly** $(\varphi_1 \vee \varphi_2)$ **Repeatedly** $\varphi_1 \vee$ **Repeatedly** φ_2

Back to Fairness

Weak fairness: An infinite execution is *fair* to a task A if, repeatedly, either A is executed or is disabled

If task is enabled, then it is eventually executed or disabled

Strong fairness: An infinite execution is *fair* to a task A, if task A is either executed repeatedly or disabled continuously from a certain step onwards

If task is repeatedly enabled, then it is repeatedly executed

Back to Fairness

Process P

nat x := 0; **bool** y := 0
A: x := x + 1
B: even(x) –> y := ¬ y

What fairness assumptions are needed so that P satisfies the spec

1. **Eventually** (x ≥ 10) :
   weak fairness for A
2. **Eventually** (y = 1) :
   strong fairness for B

Back to Fairness

Process P

nat x := 0; **bool** y := 0
A: x := x + 1
B: even(x) –> y := ¬ y

Fairness can be encoded directly in LTL!

Instead of checking if the system satisfies an LTL formula φ, check if it satisfies the formula

$$\text{FairnessAssumption} \Rightarrow \varphi$$

FairnessAssumption is an LTL formula encoding what it means for an execution to be weakly/strongly fair with respect to a task

Encoding Weak Fairness in LTL

Process P

nat x := 0; **bool** y := 0; { A, B } executed
A: x := x + 1; executed := A
B: even(x) –> y := ¬ y; executed := B

1. We add a variable called executed whose values are task names
2. Whenever a task executes, executed is assigned the name of the task

Weak fairness for a task T: An execution is weakly fair to task T if it satisfies the formula

WF_T: **Persistently** (T is enabled) ⇒ **Repeatedly** (T is executed)

Examples:

WF_A: **Repeatedly** (executed = A)

WF_B: **Persistently** (even(x)) ⇒ **Repeatedly** (executed = B)

Checking Requirements under Weak Fairness

Process P

nat x := 0; **bool** y := 0; { A, B } executed
A: x := x + 1; executed := A
B: even(x) –> y := ¬ y; executed := B

Which of the following are satisfied by P ?

1. **Eventually** ($x \geq 10$)
2. $WF_A \Rightarrow$ **Eventually** ($x \geq 10$)
3. $WF_B \Rightarrow$ **Eventually** ($y = 1$)
4. $(WF_A \wedge WF_B) \Rightarrow$ **Eventually** ($y = 1$)

What have we achieved?

- Checking if an LTL spec is satisfied under fairness assumptions is reduced to checking a modified LTL spec
- Then the verifier does not have to handle fairness explicitly

Encoding Strong Fairness

Process P

nat x := 0; **bool** y := 0; { A, B } executed
A: x := x + 1; executed := A
B: even(x) –> y := ¬ y; executed := B

Strong fairness for a task T: An infinite execution is strongly fair to task T if it satisfies the formula

SF_T: **Repeatedly** (T is enabled) ⇒ **Repeatedly** (T is executed)

Example:

SF_B: **Repeatedly** (even(x)) ⇒ **Repeatedly** (executed = B)

Note: if a spec is satisfied assuming weak fairness, then it also satisfied assuming strong fairness

Encoding Strong Fairness

Process P

nat x := 0; **bool** y := 0; { A, B } executed
A: x := x + 1; executed := A
B: even(x) –> y := ¬ y; executed := B

Strong fairness for a task T: An infinite execution is strongly fair to task T if it satisfies the formula

SF_T: **Repeatedly** (T is enabled) ⇒ **Repeatedly** (T is executed)

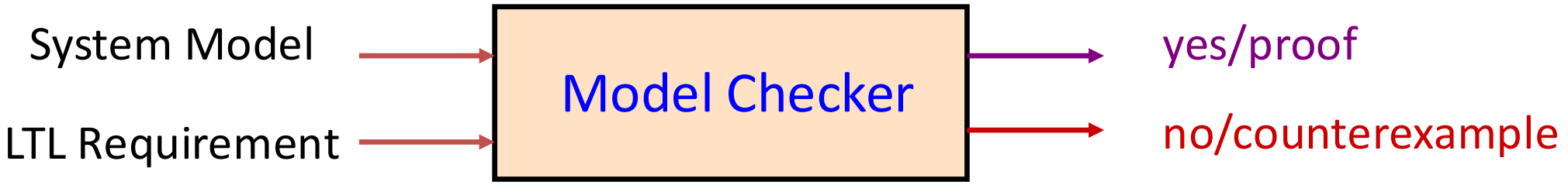
Example:

SF_B: **Repeatedly** (even(x)) ⇒ **Repeatedly** (executed = B)

Exercise: Which of the following specs are satisfied by P?

1. SF_B ⇒ **Eventually** (y = 1)
2. SF_B ⇒ **Repeatedly** (y = 1)
3. SF_B ⇒ **Persistently** (y = 1)

Model Checking



- Performed using enumerative or symbolic search through the state-space of the program
- Success story for transitioning academic research to industrial practice
- Used to debug multi-core protocols, pipelined processors, device driver code, distributed algorithms in Intel, Microsoft, IBM …

Büchi Automata

A safety monitor M classifies finite executions into good and bad

Verification of safety requirements for a component C reduces to analyzing the reachability of error states in the composition of C and M

An error execution is an execution prefix that leads the monitor to an error state

How can a monitor (aka, an automaton) classify infinite executions into good and bad?

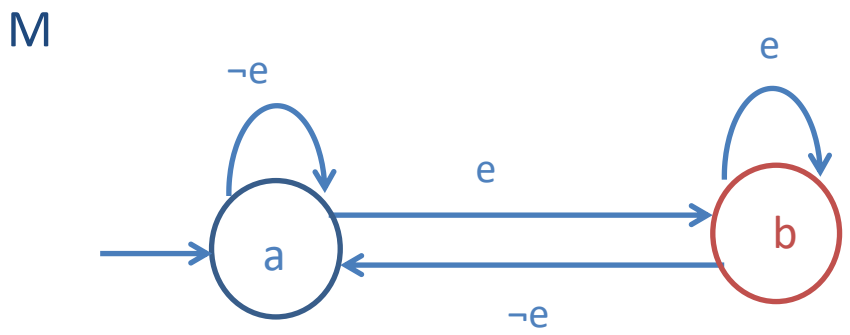
Büchi Automata

Theoretical model of Büchi automata proposed by Richard Büchi (1960)

Model checking application (1990s) using Büchi automata:

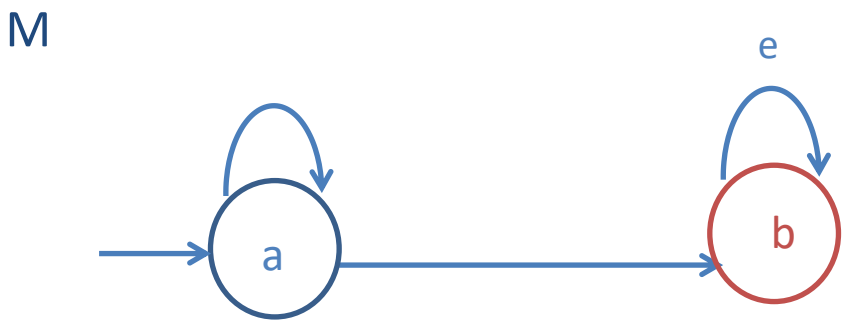
- Automatically translate LTL formula φ to a Büchi monitor M
- Consider the composition of system C and monitor M
- Reachable cycles in this composite correspond to counterexamples; if no such cycle is found, system satisfies spec
- Implemented in many model checkers (notably, SPIN)

Büchi Automaton: Example 1



- **Inputs:** Boolean variable e
- **Modes/states:** a and b; a is initial and b is accepting
- Given a trace ρ over e (i.e., infinite sequence of 0/1 values to e), there is a corresponding execution of M
- The trace ρ is accepted if accepting state is visited repeatedly
- *Language of* M = { traces in which e is satisfied repeatedly }
- M accepts ρ iff ρ ⊨ **Repeatedly** e

Büchi Automaton: Example 2



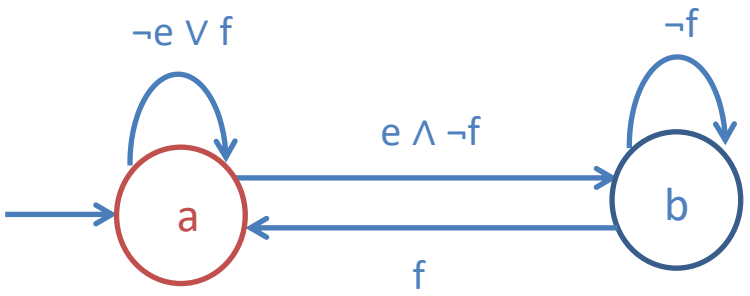
- Automaton is nondeterministic
- On a given input trace, many possible executions
- An execution is accepting if it visits accepting state repeatedly
- M accepts an input trace ρ if there exists some accepting execution on that input
- M accepts ρ iff $\rho \models$ **Persistently** e

Büchi Automaton: Example 3

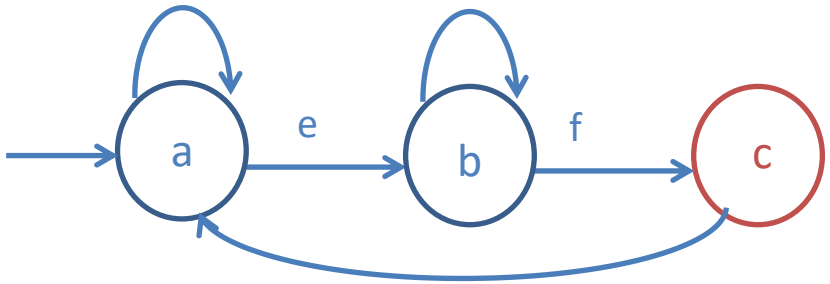
- Design a Büchi automaton M such that

  M accepts trace ρ iff ρ ⊨ **Always** (e ⇒ **Eventually** f)

- **Inputs:** Boolean values for e and f
- In an accepting execution, every e must be eventually followed by an f



Büchi Automaton: Example 4



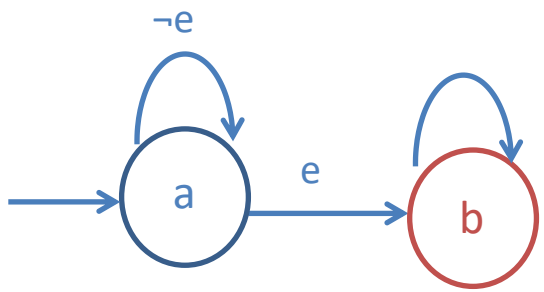
Which traces does this accept? Express it in LTL

M accepts ρ iff $\rho \models$ **Repeatedly** e $\wedge$ **Repeatedly** f

Büchi Automaton M Definition

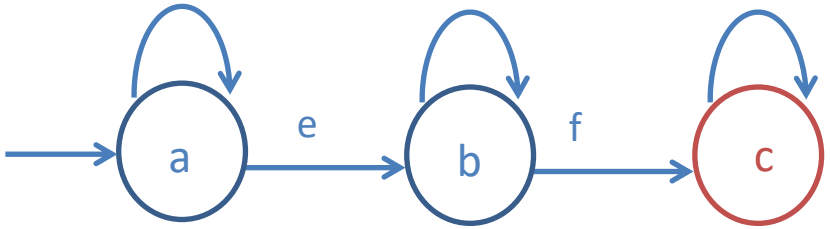
- Set of Boolean V of input variables
- Finite set Q of states
- Set Init of initial states
- Set F of accepting states
- Set of transitions of the form $q \text{ –G–> } q'$ where G is a Boolean expression over V
- Given an input trace $\rho = v_1, v_2, v_3, \ldots$ over V,
  an *accepting execution* of M over ρ is an infinite sequence of states $q_0, q_1, q_2, \ldots$
  where:
  - q_0 is initial
  - For each i, there is an edge $q_i \text{ –G–> } q_{i+1}$ such that input v_i satisfies G
  - There are infinitely many positions i such that state q_i is in F
- M *accepts* input trace ρ if there is an accepting execution of M over ρ

Büchi Automata: More Examples



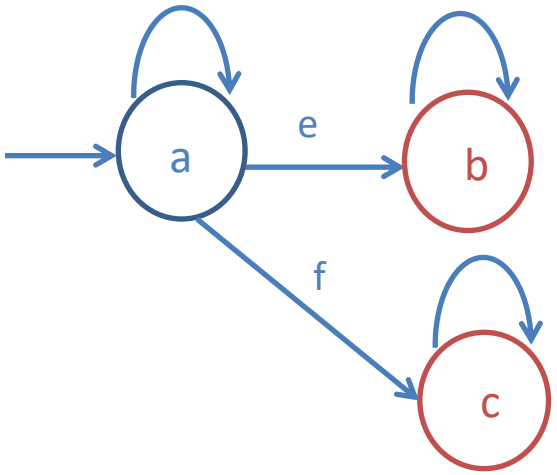
Eventually e

Büchi Automata Examples

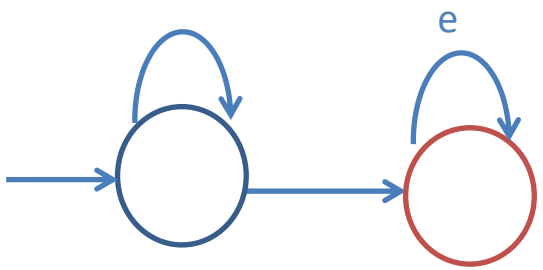


Eventually [e ∧ **Next Eventually** f]

Eventually e ∨ **Eventually** f



Nondeterministic Büchi Automaton

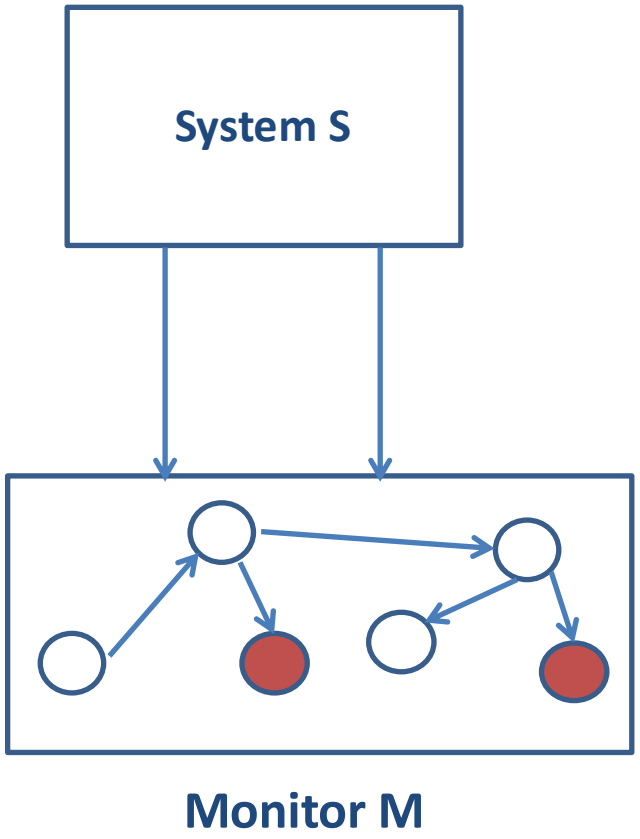


Persistently e

Can we construct an equivalent deterministic Büchi automaton?

No! Non-determinism is sometimes necessary!

Safety Monitors



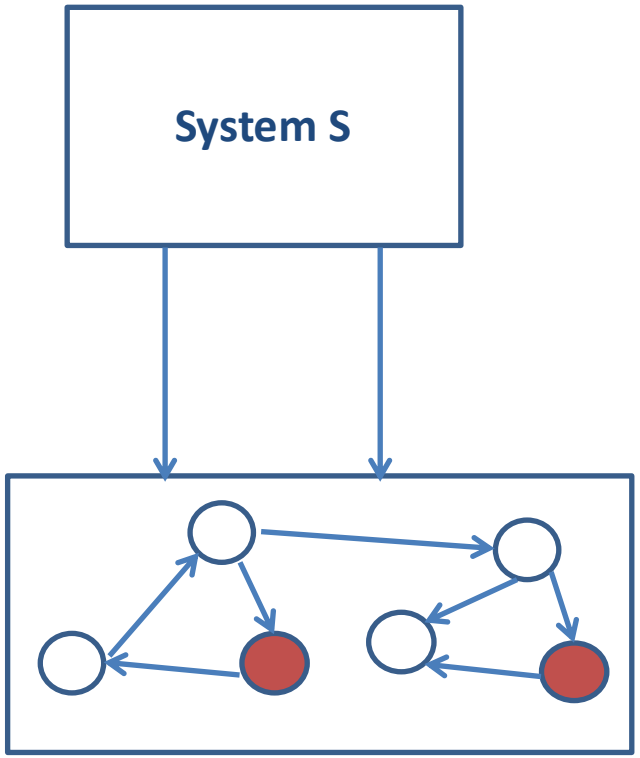
Is there an execution of **S** that makes **M** enters an error state?

M is designed so that such an execution indicates a bug!

Verification reduces to reachability

Check if an error state is reachable in composition of **S** and **M**

Büchi Monitors



Büchi Monitor M

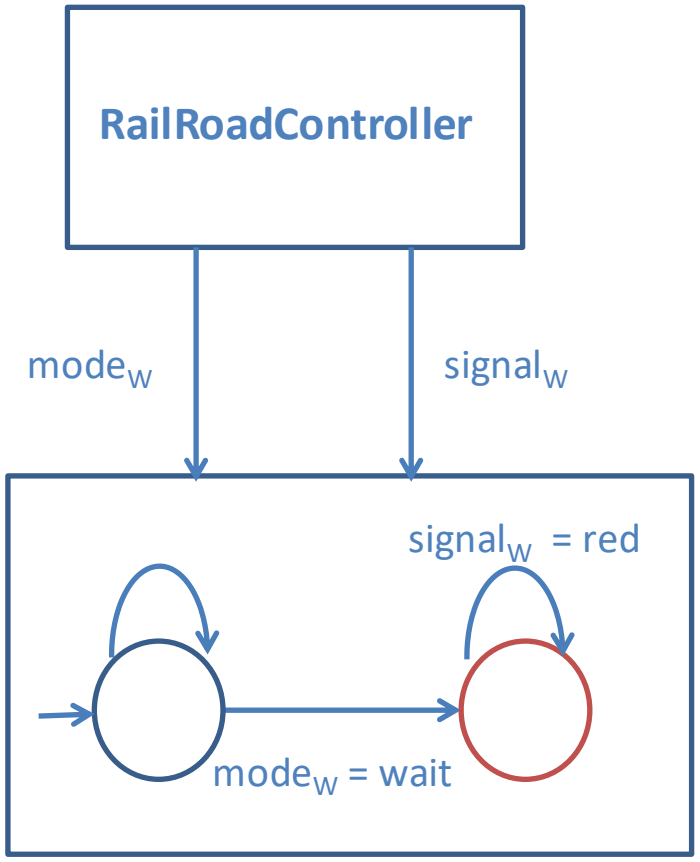
Is there an execution of **S** which is accepted by **M**? (i.e., an execution in which some error state of **M** appears repeatedly?)

M is designed so that such an execution indicates a bug!

Verification reduces to search for cycles

Check if there is a reachable cycle containing an error state in the composition of **S** and **M**

Example Büchi Monitor



Büchi Monitor M

Correctness requirement:

Always [(West train is waiting) ⇒
Eventually (West signal is green)]

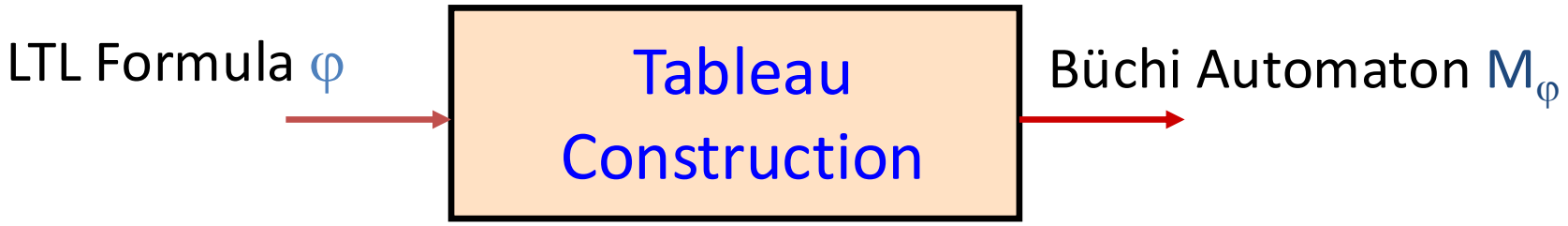
Requirement violation:

Infinite execution where, at some step, west train is waiting and in all subsequent steps west signal is red

Verification:

Search for reachable cycle containing red monitor state in the combined system

From LTL to Büchi Automata



Automaton M_φ accepts exactly those traces that satisfy formula φ

To check if a system C satisfies the LTL requirement φ:

1. construct the Büchi automaton M_φ corresponding to φ
2. composition C with M_φ
3. search for cycles in composition

Tableau Construction Example

Consider **Always** e ∧ **Eventually** f: **A** e ∧ **E** f



A state is a collection of formulas that must be satisfied

Initial state contains given formula

Formulas in a state q must be consistent with rules of logical connectives: for example:

- if q has $\varphi_1 \wedge \varphi_2$, then it must have both φ_1 and φ_2
- if q has $\varphi_1 \vee \varphi_2$, then it must have at least one of φ_1 and φ_2
- If q has $\neg\varphi$, then it does not have φ, and vice versa

Omega-Regular Languages

- The *language* of a Büchi automaton is the set of traces it accepts
- Such languages are called *ω-regular*

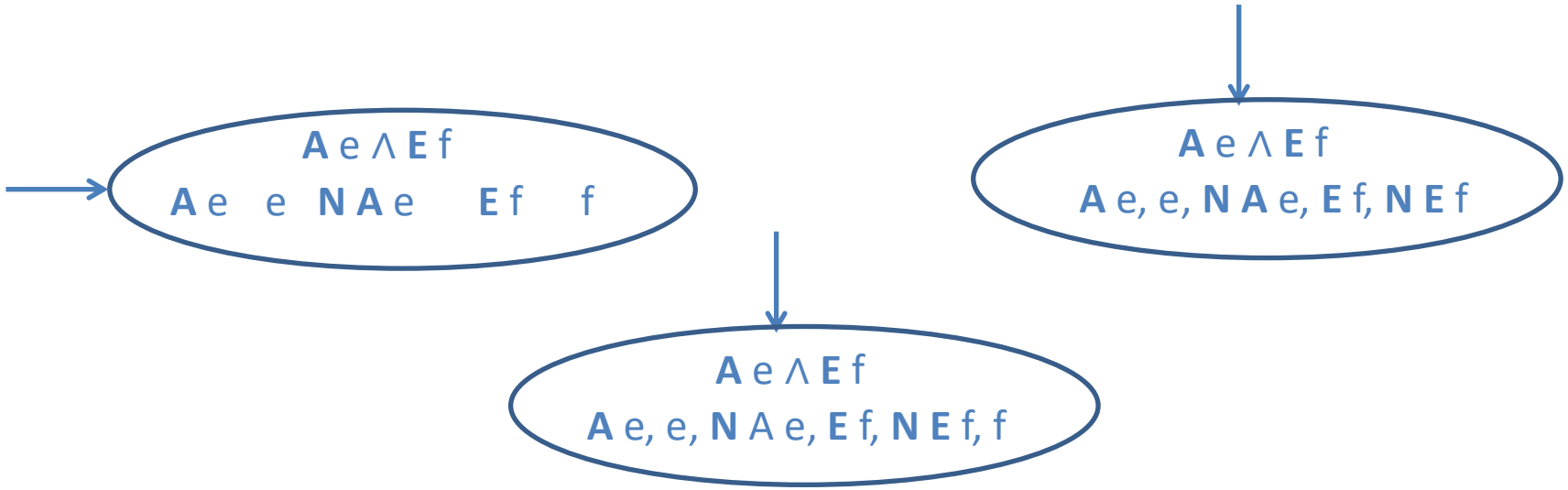
- There is a well-developed theory of ω-regular languages

- It is analogous the classical theory of regular languages
  (i.e., languages of finite strings of input characters accepted by finite automata)

- **Relevance to us:** Given an LTL formula φ, there is an algorithm to construct a Büchi automaton M_φ that accepts exactly the traces that satisfy φ

Tableau Construction Example

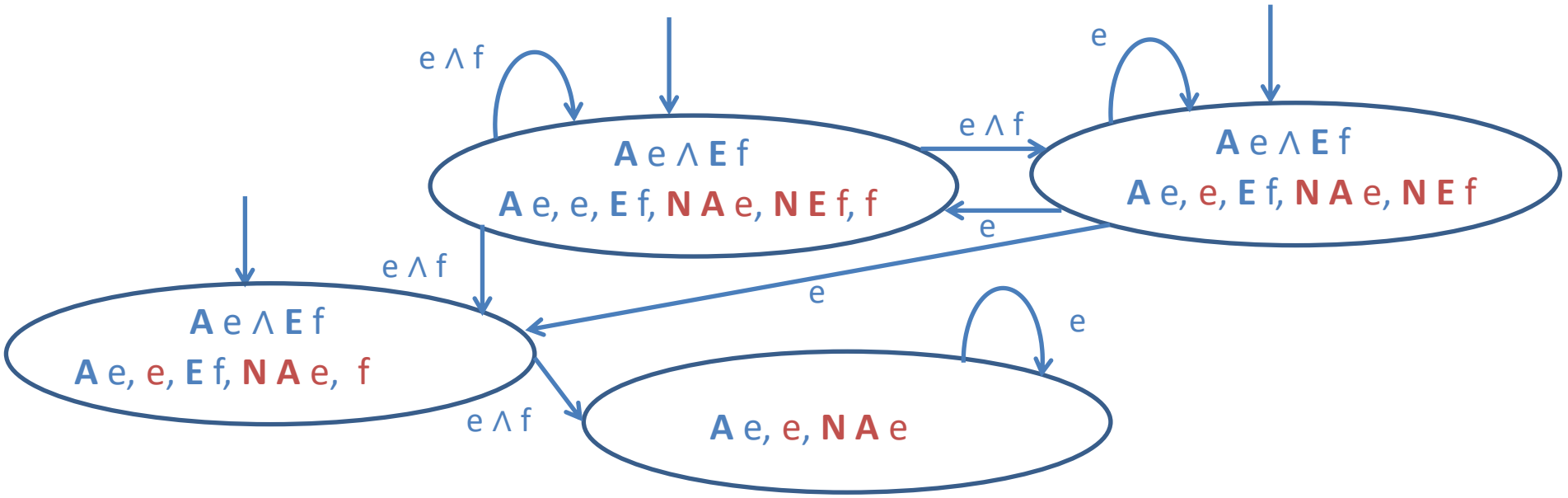
Consider **Always** e ∧ **Eventually** f: **A** e ∧ **E** f



- If a state has **Always** φ, it must have both φ and **Next Always** φ
- If a state has **Eventually** φ, it must have either φ or **Next Eventually** φ or both
  This leads to 3 cases

Tableau Construction Example

Consider **Always** e ∧ **Eventually** f: **A** e ∧ **E** f

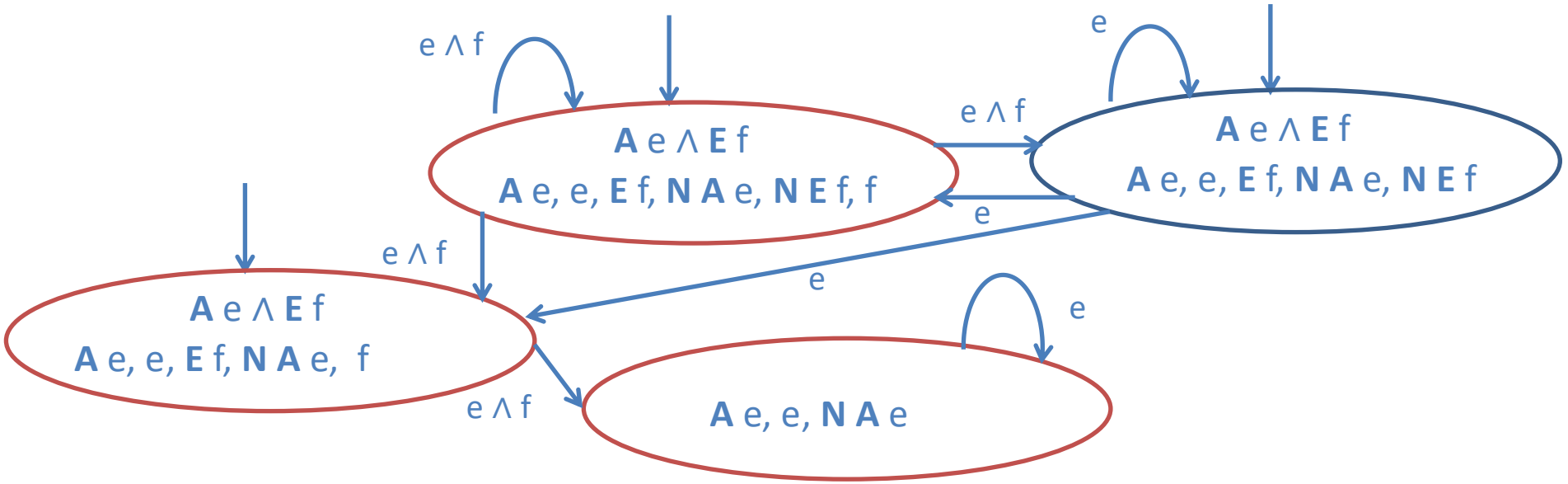


Transition Rules:

1. If a state contains **Next** φ then add transition to each state containing φ
2. If a state contains base formula φ, then φ must hold on outgoing transitions

Tableau Construction Example

Consider **Always** e ∧ **Eventually** f: **A** e ∧ **E** f

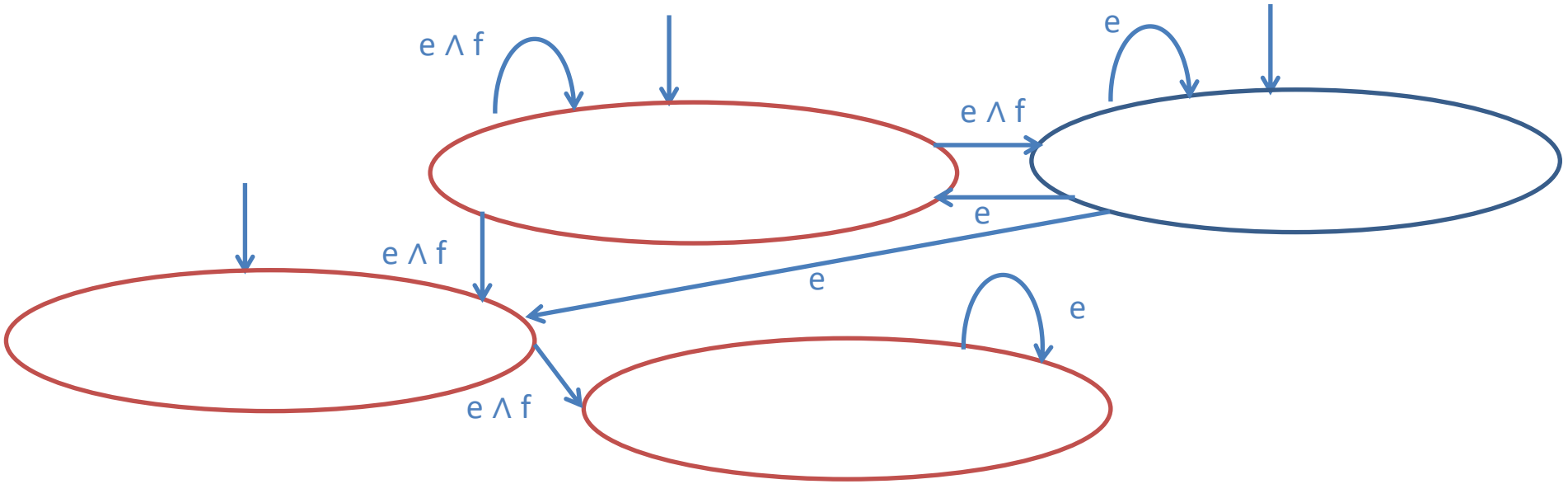


Acceptance condition: Satisfaction of eventuality should not be postponed forever

Accepting states: States that either contain f or do not contain **E** f

Tableau Construction Example

Consider **Always** e ∧ **Eventually** f: **A** e ∧ **E** f

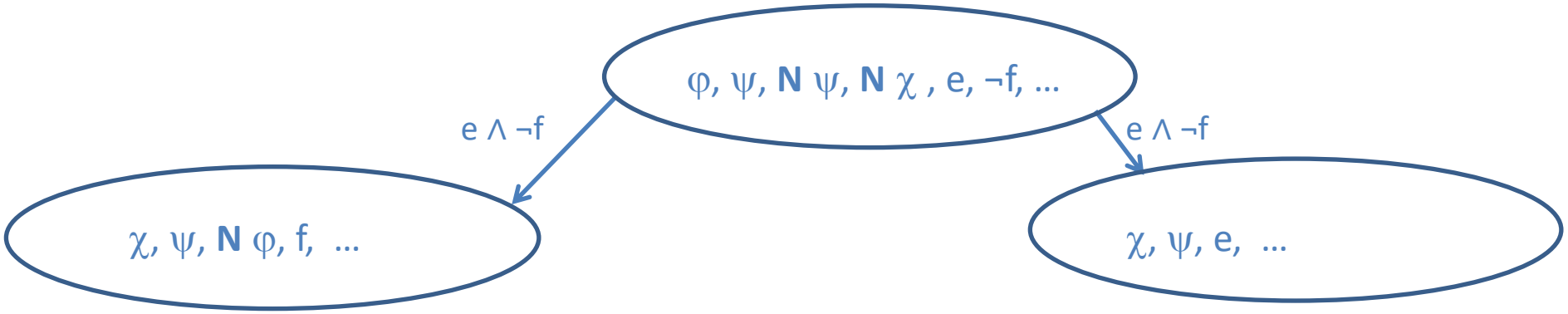


Indeed, this is a correct Büchi automaton for the given formula!

Tableau Construction Overview

Automaton/tableau state: Collection of relevant LTL formulas

Intended meaning: All the formulas in a state q must hold on every infinite path starting at q



Local consistency rules ensure that for every non-atomic formula φ, the state contains additional formulas ensuring that φ holds

Transition rules ensure that

1. every atomic formula holds at current time, and
2. all Next formulas are propagated to next state

Formal Construction

Given an LTL-formula φ, define set Sub(φ), the *closure* of φ

Sub(φ) consists of formulas that are relevant to evaluation of φ:

- It contains all the subformulas of φ
- If it contains **Always** ψ, it also contains **Next Always** ψ
- If it contains **Eventually** ψ, it also contains **Next Eventually** ψ
- If it contains φ_1 **Until** φ_2, it also contains **Next** (φ_1 **Until** φ_2)

Example:

Sub(**Always Eventually** e $\wedge$ **Next** f) =

{ **Always Eventually** e $\wedge$ **Next** f,
Always Eventually e, **Eventually** e, e
Next f, f,
Next Eventually e, **Next Always Eventually** e }

Note: Size of Sub(φ) is linear in the size of φ

Tableau States

A state of the desired automaton is a subset of Sub(φ) that satisfies some consistency rules:

- Does not contain both a formula φ_1 and its negation $\neg\varphi_1$
- Contains $\varphi_1 \wedge \varphi_2$ exactly when it contains both φ_1 and φ_2
- Contains $\varphi_1 \vee \varphi_2$ exactly when it contains at least one of φ_1 and φ_2
- Contains $\varphi_1 \Rightarrow \varphi_2$ exactly when it contains at least one of $\neg\varphi_1$ and φ_2
- If it contains **Always** φ_1 then it contains both φ_1 and **Next Always** φ_1
- If it contains **Eventually** φ_1 then it contains at least one of φ_1 and **Next Eventually** φ_1
- If it contains φ_1 **Until** φ_2 them it contains φ_1 or both φ_2 and **Next** (φ_2 **Until** φ_2)

Note: Number of possible states is exponential in the size of φ

Example Construction

Formula φ = **Eventually** e $\wedge$ **Next** $\neg$e

Sub(φ) = { **E** e $\wedge$ **N** $\neg$e, **E** e, e, **N** $\neg$e, $\neg$e, **N E** e }

Tableau states:

q_0 = { e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_1 = { e, **N E** e, **E** e }

q_2 = { e, **N** $\neg$e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_3 = { e, **E** e }

q_4 = { $\neg$e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_5 = { $\neg$e, **N E** e, **E** e }

q_6 = { $\neg$e, **N** $\neg$e }

q_7 = { $\neg$e }

Tableau Construction Continued

Input variables V**:** base formulas appearing in φ

States: Consistent subsets of Sub(φ)

Initial states: States that contain the formula φ

Transitions: q –G–> q’ is a transition provided

- **Next** ψ is in q exactly when ψ is in q’
- If a base formula e is in q, then e is a conjunct in G, else ¬e is a conjunct in G

Example Construction Continued

Formula φ = **Eventually** e $\wedge$ **Next** $\neg$e

Tableau states:

q_0 = { e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_1 = { e, **N E** e, **E** e }

q_2 = { e, **N** $\neg$e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_3 = { e, **E** e }

q_4 = { $\neg$e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_5 = { $\neg$e, **N** E e, **E** e }

q_6 = { $\neg$e, **N** $\neg$e }

q_7 = { $\neg$e }

Transitions from q_0:

q_0 –e–> q_4

q_0 –e–> q_5

Transitions from q_1:

q_1 –e–> q_0

q_1 –e–> q_1

q_1 –e–> q_2

q_1 –e–> q_3

Transitions from q_6:

q_6 –($\neg$e)–> q_6

q_6 –($\neg$e)–> q_7

Tableau Construction: Acceptance

For a subformula **Eventually** ψ, need to ensure that satisfaction of ψ is not postponed forever.

Whenever **Eventually** ψ appears is in a state either ψ or **Next Eventually** ψ, or both, are included

Define F to be the set of tableau states that either include ψ or exclude **Eventually** ψ

Accepting condition: Repeatedly F

Similarly, for a subformula **Always** ψ,

1. Define F' to be the set of states that either include **Always** ψ or exclude ψ
2. A state in F' is required to appear repeatedly on an accepting run

Example Construction Continued

Formula φ = **Eventually** e $\wedge$ **Next** $\neg$e

Tableau states:

q_0 = { e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_1 = { e, **N E** e, **E** e }

q_2 = { e, **N** $\neg$e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_3 = { e, **E** e }

q_4 = { $\neg$e, **N** $\neg$e, **N E** e, **E** e, **E** e $\wedge$ **N** $\neg$e }

q_5 = { $\neg$e, **N E** e, E e }

q_6 = { $\neg$e, **N** $\neg$e }

q_7 = { $\neg$e }

Accepting states:

{ q_0, q_1, q_2, q_3, q_6, q_7 }

Initial states: { q_0, q_2, q_4 }

Transitions from q_0:

q_0 –e–> q_4

q_0 –e–> q_5

Transitions from q_1:

q_1 –e–> q_0

q_1 –e–> q_1

q_1 –e–> q_2

q_1 –e–> q_3

Transitions from q_6:

q_6 –(¬e)–> q_6

q_6 –(¬e)–> q_7

Handling Acceptance

In general, if there are multiple temporal formulas, the acceptance condition should ensure that each is satisfied

Generalized Büchi Automaton: Modest syntactic generalization

Automaton M has k accepting sets F_1, F_2, … F_k

An execution is accepting if for each j, some state in F_j appears repeatedly

Repeatedly $F_1 \wedge$ **Repeatedly** $F_2 \wedge \ldots \wedge$ **Repeatedly** F_k

- It is possible to *compile* a generalized Büchi automaton to a standard Büchi automaton
- It is also possible to adapt cycle-detection algorithms to handle multiple accepting sets

Tableau Construction: Summary

Correctness: A trace over V satisfies a given LTL formula φ iff it is accepted by the Generalized Büchi Automaton M_φ

Complexity: Size of M_φ is 2^n, where n is the size of φ (such a blow-up is unavoidable)

Practical implementations with a number of optimizations exist

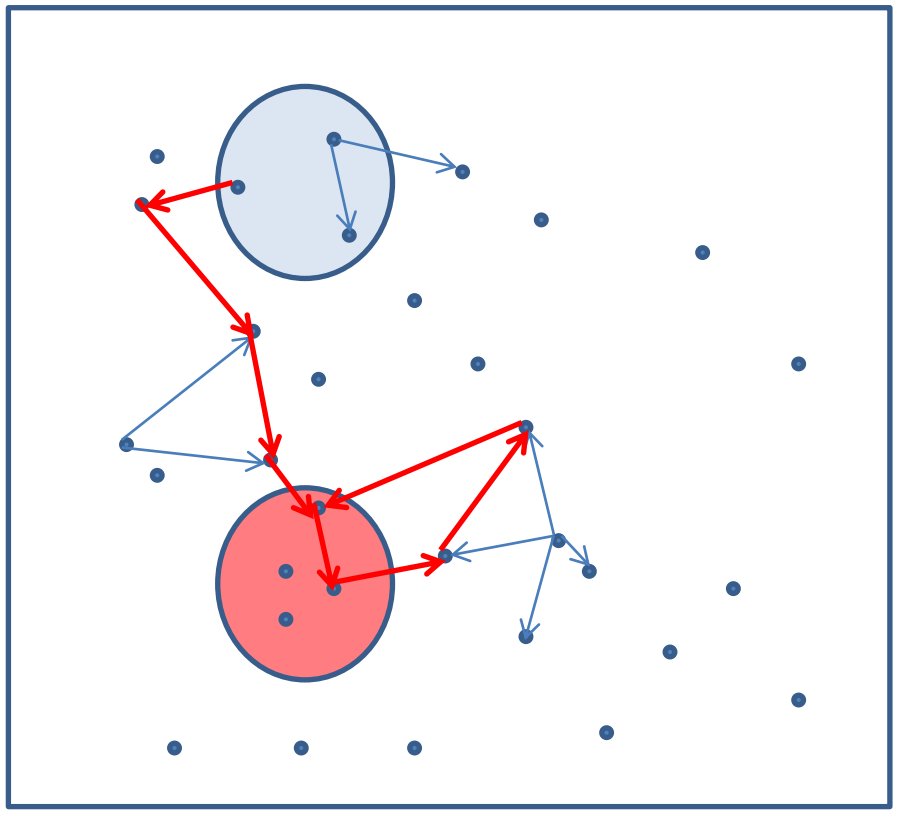
Reachability Problem for Transition Systems



- Is there a (finite) execution from an initial state to a state satisfying φ
- Checking whether φ is an invariant of T reduces to checking if ¬φ is unreachable
- Verification techniques
  1. Proof-based (inductive invariants)
  2. Enumerative on-the-fly search (see notes)
  3. Symbolic search based on iterative image computation

Repeatability Property for Transition Systems

Transition System = (States, Initial states, Transitions)



Property φ : Subset of states

Property φ is *repeatable* if there exists an infinite execution that satisfies **Repeatedly** φ

Is there a state s such that
1. s is reachable
2. s satisfies φ
3. there is a cycle containing s

Repeatability Problem for Transition Systems



Is there an infinite execution along which states satisfying φ appear repeatedly?

To check whether a system C satisfies an LTL formula φ, check if property Mode is accepting is repeatable in composition of C and Büchi monitor $M_{\neg\varphi}$

Verification techniques (not covered, see Chap 5)

1. **Proof-based:** Ranking functions
2. **Enumerative:** Nested Depth-first Search
3. **Symbolic search**

Credits

Notes based on Chapter 5 of

Principles of Cyber-Physical Systems
by Rajeev Alur
MIT Press, 2015